BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 063 · Lesson 063 · Thinking Logically: Conditions and Decisions

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What does a Boolean expression evaluate to?

Q2. What is a decision boundary?

Q3. Why is condition order important in else-if?

Q4. 029

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

predicate — Boolean condition.

selection — Branch based on a condition.

boundary test — Input near comparison threshold.

short-circuit — Skipping redundant Boolean evaluation.

New learning

Identify — Identify decisions in the solution and write precise Boolean conditions, including valid-data boundaries.

Model — Order conditions so cases are mutually understood: mark>=70 must be considered before mark>=40.

Evaluate — Use truth tables or decision tables to test normal, erroneous and boundary input values.

Worked example

Trace Grade(40) using the conditions shown.

1. Explain the decision — Check comparisons in order; stop after first matching return.

We Do · Practise with guidance

Trace Grade(105). Why should validity be checked first?

```
105>100, so return Invalid before the threshold branches.
```

Explain the decision

Without range check, 105 would incorrectly be classified High.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Trace Grade for marks 39,40,69,70 and give output at each. [4]

Question 2

Write a Boolean rule for a valid age 11–18 inclusive, and give two boundary tests. [6]

Question 3

A program checks if(mark>=40) before else-if(mark>=70), then outputs Pass for 85. Explain bug and repair. [6]

Exit Ticket · Answer before checking

With else-if, a later condition always runs even if an earlier branch matches.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. True or False.

Q2. A value at which condition outcome changes.

Q3. Earlier branch can prevent later branch ever being checked.

Q4. Simplify (A∧B)∨(A∧¬B), showing one law.

You Do pages · Independent questions

Q1. 39 Below pass [1],40 Pass [1],69 Pass [1],70 High [1].

Q2. age>=11 && age<=18 [3]; test 11 valid [1],18 valid [1],10 or 19 invalid [1].

Q3. 85 meets first condition [2], second branch never reached [1], test >=70 first [2], 85 then High [1].

Exit Ticket · Model answer

False. Later branches are skipped after a matching branch.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: